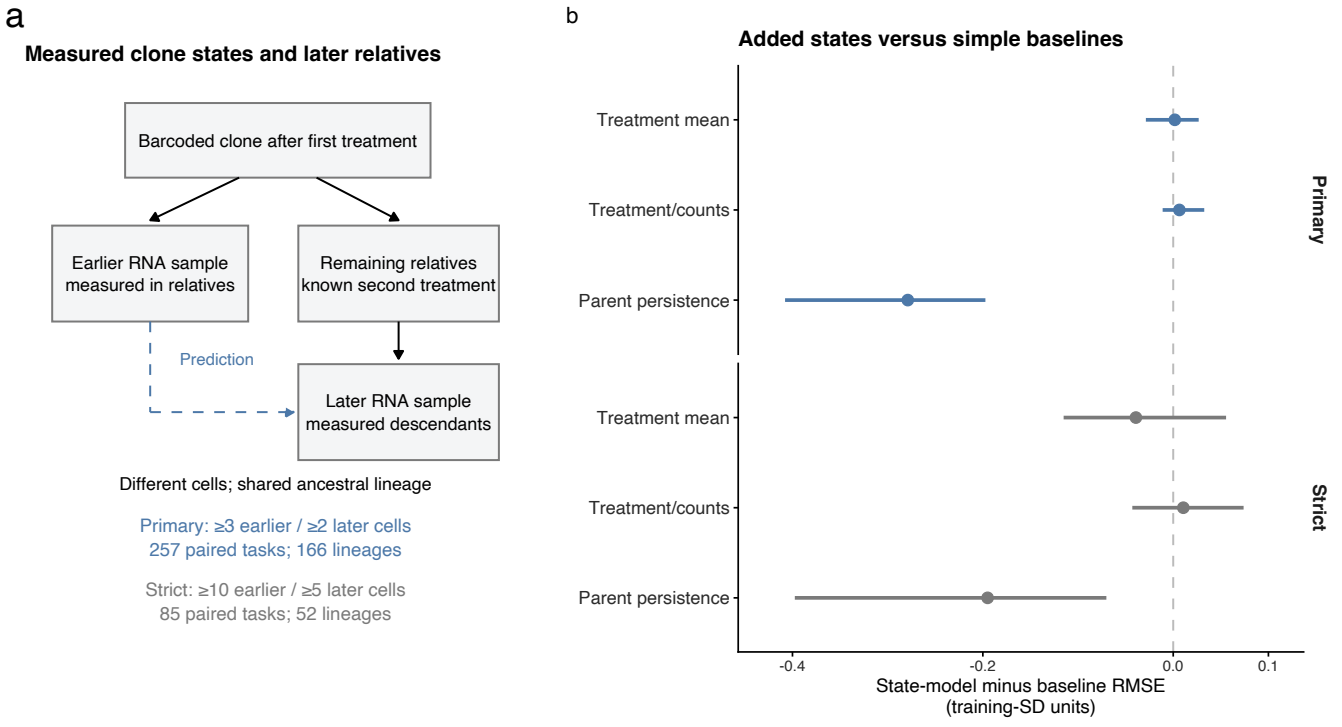
a
Measured clone states and later relatives
Barcoded clone after first treatment
Earlier RNA sample measured in relatives
Remaining relatives known second treatment
Prediction
Later RNA sample measured descendants
Different cells; shared ancestral lineage
Primary: ≥3 earlier / ≥2 later cells
257 paired tasks; 166 lineages
Strict: ≥10 earlier / ≥5 later cells
85 paired tasks; 52 lineages
b
Added states versus simple baselines
Treatment mean
Treatment/counts
Parent persistence
Primary
Treatment mean
Treatment/counts
Parent persistence
Strict
-0.4
-0.2
0.0
0.1
State-model minus baseline RMSE
(training-SD units)